

Heterogeneity in 2,6-Linked Sialic Acids Potentiates Invasion of Breast Cancer Epithelia

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Cite This: *ACS Cent. Sci.* 2021, 7, 110–125



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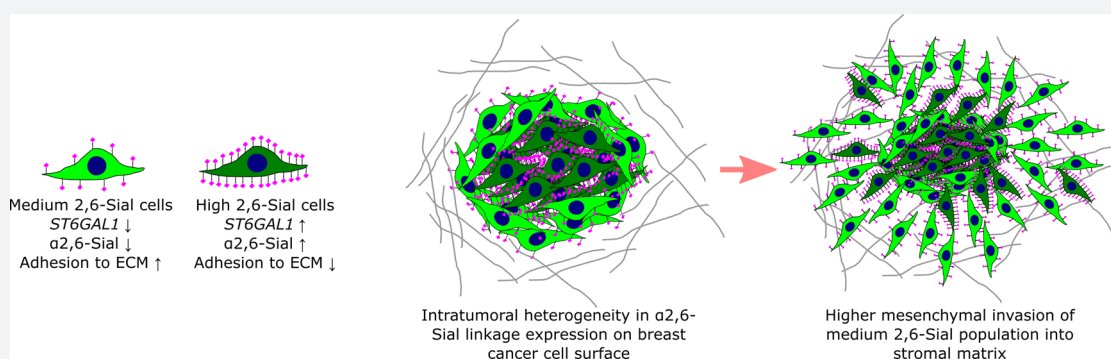
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ABSTRACT: Heterogeneity in phenotypes of malignantly transformed cells and aberrant glycan expression on their surface are two prominent hallmarks of cancers that have hitherto not been linked to each other. In this paper, we identify differential levels of a specific glycan linkage: α 2,6-linked sialic acids within breast cancer cells *in vivo* and in culture. Upon sorting out two populations with moderate, and relatively higher, cell surface α 2,6-linked sialic acid levels from the triple-negative breast cancer cell line MDA-MB-231, both populations (denoted as medium and high 2,6-Sial cells, respectively) stably retained their levels in early passages. Upon continuous culturing, medium 2,6-Sial cells recapitulated the heterogeneity of the unsorted line whereas high 2,6-Sial cells showed no such tendency. Compared with high 2,6-Sial cells, the medium 2,6-Sial counterparts showed greater adhesion to reconstituted extracellular matrices (ECMs) and invaded faster as single cells. The level of α 2,6-linked sialic acids in the two sublines was found to be consistent with the expression of a specific glycosyl transferase, *ST6GAL1*. Stably knocking down *ST6GAL1* in the high 2,6-Sial cells enhanced their invasiveness. When cultured together, medium 2,6-Sial cells differentially migrated to the edge of growing tumoroid-like cocultures, whereas high 2,6-Sial cells formed the central bulk. Multiscale simulations in a Cellular Potts model-based computational environment calibrated to our experimental findings suggest that differential levels of cell–ECM adhesion, likely regulated by α 2,6-linked sialic acids, facilitate niches of highly invasive cells to efficiently migrate centrifugally as the invasive front of a malignant breast tumor.

INTRODUCTION

One of the hallmarks of malignant tumors is heterogeneity in the phenotypes of its constituent transformed epithelia. Observations of phenotypic heterogeneity can be traced back to the demonstration by Hawkins and co-workers of variable expression of the estrogen receptor (ER) among cells within a tumor. With time, evidence of intratumoral variation in expression was discovered for several genes/markers and is responsible for determining clinical behavior and response to treatment.^{1–5} Intratumoral heterogeneity can also contribute to misdiagnosis of the aggressiveness and grade of breast cancer leading to its mistreatment.^{6–8}

A combination of genomic and epigenomic aberrations and loss in a dynamic and reciprocal regulation of homeostasis by the tissue microenvironment and stochasticity leads to diversity in protein expression, localization, and interaction within cells

belonging to the same population. This diversity in turn leads to heterogeneity in cellular phenotypes.^{9–12} However, proteins are not the only molecular species to show such alterations in malignant contexts. Changes in levels of sugars on the surfaces of cancer cells, when compared with their untransformed counterparts, have been demonstrated for a long time.^{13–15} Further studies show that altered levels of *N*- and *O*-linked glycosylations in transformed epithelia and tumor-associated stromal cells impact the progression and metastasis of cancer as well as its

Received: May 11, 2020

Published: January 4, 2021



ACS Publications

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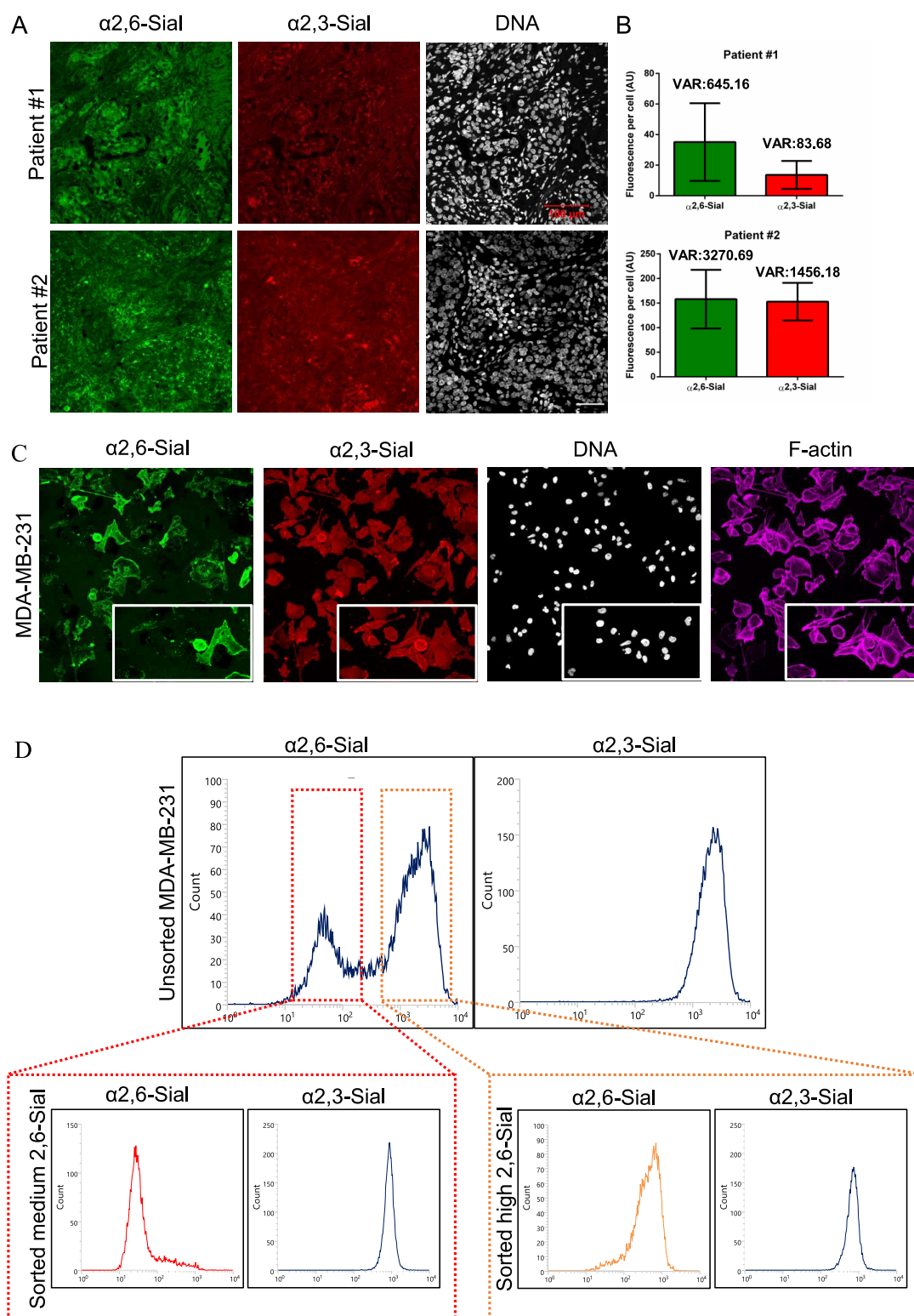


Figure 1. $\alpha 2,6$ -Sialic acid heterogeneity in breast cancer. (A) Confocal micrographs showing $\alpha 2,6$ -sialic acid (SNA-FITC, green) and $\alpha 2,3$ -sialic acid (MAA-TRITC, red) staining of breast cancer sections from two individuals (top and bottom rows) showing heterogeneity in $\alpha 2,6$ -sialic acid linkage expression. The nucleus is stained with DAPI (white) (scale bar: 100 μm). (B) Bar graphs showing quantification of individual sialic acid levels from breast cancer sections shown in part A. Error bars represent mean $\pm$ SD. (C) Confocal micrographs showing heterogeneity in $\alpha 2,6$ -Sial levels (SNA-FITC green) and uniform $\alpha 2,3$ -sialic levels (MAA-TRITC, red) in invasive breast cancer cell line MDA-MB-231 using lectin cytochemical fluorescence. Cells are counterstained for nucleus with DAPI (white) and F-actin with phalloidin (Magenta). Insets of a subfield within the images shown in bottom right corner. (D) Lectin-based flow cytometry profiles of MDA-MB-231 cells show bimodal distribution of $\alpha 2,6$ -Sial levels (top left) and unimodal distribution of $\alpha 2,3$ -Sial levels (top right). Red inset shows moderate levels of $\alpha 2,6$ -Sial (left) and unchanged $\alpha 2,3$ -Sial (right) in sorted medium $\alpha 2,6$ -Sial cells. Orange inset shows higher $\alpha 2,6$ -Sial (left) and unchanged $\alpha 2,3$ -Sial (right) levels in sorted high $\alpha 2,6$ -Sial cells.

response to chemotherapy.^{16–18} For example, aberrantly glycosylated β 1-integrin leads to altered cell–ECM adhesion, thereby aiding cancer cell invasion and metastasis.¹⁹ An increase in Sialyl Lewis^x enhances adhesion of cancer cells to endothelial cells via selectins, leading to colonization of distant organs.²⁰ O-GlcNAcylated c-Myc can compete with its unglycosylated counterpart in phosphorylation leading to increased stability and thereby increase cancer cell proliferation.²¹ Hypersialylation is one of the frequently observed changes in cell surface glycosylation seen in many cancer types.^{22,23} Selective enrichment of terminal α 2,6-linked sialic acids (referred to here onward as α 2,6-Sial), due to overexpression of *ST6GAL1*, in cancer cell glycocalyx can elicit a wide range of biological outcomes like protection from hypoxia, resistance to chemotherapy, prosurvival and conferral of cancer stem cell phenotype.^{24–27}

To the best of our knowledge, there is no literature on differential levels of glycans on the surface of transformed cells within growing tumors. In the present study, we investigated this question in the context of sialic acid expression in breast cancer. Using a combination of lectin-based flow cytometry, cytochemistry, and glycomics using mass spectrometry, we demonstrated diverse expression profiles of α 2,6-Sial within breast cancer cells *in vitro* and *in vivo*. The diversity was glycan-specific: α 2,3-Sial or other assessed oligosaccharides did not show such differential expression. We found two distinct cell populations with moderate and high levels of α 2,6-Sial in the triple-negative breast cancer cell line MDA-MB-231. Combining cell biological assays with agent-based modeling simulations, we demonstrate how the distinct glycan levels result in differential migration of more invasive epithelia to the invading edge of cultured 3D tumoroids. A better understanding of intratumor glycobiological heterogeneity is certain to impact breast cancer diagnosis and treatment in the future.

■ RESULTS

Breast Cancer Epithelia Show Differences in α 2,6-Sial Levels. We assayed for intercellular differences in α 2,6- and α 2,3-Sial levels in breast cancer sections using lectin histochemistry. FITC-conjugated *Sambucus nigra* (SNA) lectin and TRITC-conjugated *Maackia amurensis* (MAA) lectin were used as probes for α 2,6- and α 2,3-Sial, respectively. Examination of tumor sections from 6 of 8 patients showed signals for both sugar linkages when compared to appropriate controls (staining from 2 representative patients shown in Figure S1 and Figure 1). However, although cells in the sections stained uniformly for α 2,3-Sial (Figure 1A, red), cellular staining for α 2,6-Sial (Figure 1A, green) was variegated: rounded patches of cells with high levels of α 2,6-Sial were surrounded by dispersed populations with comparatively lower levels (Figure 1A). This was confirmed through per-cell quantification of cancer cells that revealed a greater variance in cell-specific expression of α 2,6-Sial relative to α 2,3-Sial (Figure 1B). Whereas elevated levels of α 2,6-Sial in breast cancer epithelia have been previously reported,^{28,29} our report is the first to document intercellular diversity of expression of a specific sugar linkage (α 2,6-Sial) *in vivo*.

We next asked whether such heterogeneous expression could also be observed within cultured breast cancer cell lines. Lectin cytochemistry of the breast cancer cell line MDA-MB-231, using FITC-conjugated SNA and TRITC-conjugated MAA, revealed higher signals (Figure 1C) compared with lactose sugar controls (Figure S2). Similar to our *in vivo* findings, we observed marked

variation in α 2,6-Sial linkage levels between MDA-MB-231 cells in the same field (Figure 1C, green, inset). Such variations were not appreciable for α 2,3-Sial levels between the same cells (Figure 1C, red, inset). To probe the distinction in sialic acid levels at single-cell resolution (Figure S3), we combined lectin-binding with flow cytometry and were able to discern two subpopulations of MDA-MB-231 cells with distinct levels of α 2,6-Sial, evident from a bimodal distribution of the staining intensity histogram [unstained cells or cells stained with FITC were used as negative control (Figures S4 and S5) (a third subpopulation is a minor fraction that does not stain for α 2,6-Sial; Figure 1D, left, and Figure S6)]. The level of α 2,3-Sial in the same population showed a sharp unimodal distribution (Figure 1D, right) confirming our observation from Figure 1C. We confirmed the specificity of lectins used in the study by treating the unsorted MDA-MB-231 cells with sialidase from *Clostridium perfringens*, which preferentially cleaves α 2,3-Sial. We observed a decrease in α 2,3-Sial levels upon treatment but not in α 2,6-Sial levels (Figure S7). A wide distribution of α 2,6-Sial levels and a comparatively sharper expression of α 2,3-Sial levels were also observed for the metastatic triple-negative BT-549 breast cancer cells (Figure S8).

We then sorted these subpopulations based on α 2,6-Sial as shown in Figure 1D and will refer to them here onward as low 2,6-Sial (cells that did not stain for α 2,6-Sial), medium 2,6-Sial, and high 2,6-Sial cells. After sorting, the individual populations were cultured separately. In early passages, the medium 2,6-Sial subpopulation continued to show a sharp and unimodal peak of moderate α 2,6-Sial levels coincident with the first peak of the bimodal distribution seen in unsorted MDA-MB-231 cells (Figure 1D, red inset, left). Early-passage high 2,6-Sial cells also showed a unimodal peak of α 2,6-Sial coincident with the second peak of the bimodal distribution seen in the unsorted MDA-MB-231 cells (Figure 1D, orange inset, right). To our surprise, low 2,6-Sial cells, upon culture, shifted to stably express α 2,6-Sial to levels concurrent with the medium 2,6-Sial subpopulation (Figure S6, green inset, left). All of the three subpopulations showed similar levels of α 2,3-Sial (Figure 1D, red inset, orange inset, right; and Figure S6, green inset, right).

Next, we confirmed the differential levels of α 2,6-Sial in the sorted, early-passage cultures of the high and medium 2,6-Sial cells using lectin cytochemical fluorescence (Figure S9, green), even though α 2,3-Sial levels remained unchanged between them (Figure S9, red). We also probed for other glycans in unsorted MDA-MB-231 cells. Levels of bisecting, bi- (Figure S10, green, top row), tri-, and tetra-antennary N-linked glycans (Figure S10, green, middle row) as well as Core 1, mono/di sialyl Core 1 O-linked glycans (Figure S10, green bottom row) showed uniform expression. Similarly, sharp unimodal distribution was observed when the sorted high and medium 2,6-Sial cells were probed for the above glycans (Figure S11). Furthermore, we confirmed the existence of α 2,6-Sial diversity using mass spectrometry. Released glycans from cell lysate (N-, O- and glycosphingolipid (GSL)-glycans) were identified and characterized using well-established porous graphitized carbon liquid chromatography (PGC-LC) and negative mode electrospray ionization tandem mass spectrometry (ESI-MS/MS) analysis.^{30,31} Representative extracted ion chromatograms (EICs) of sialylated structures from N-, O-, and GSL-glycans are shown in Figures S12 and S13, respectively. The EIC of monosialylated (m/z 965.8²⁻ and m/z 1038.8²⁻) and disialylated biantennary structures (m/z 1111.4²⁻ and m/z 1184.4²⁻) of the three populations: unsorted, high, and medium 2,6-Sial showed that the relative abundance of the

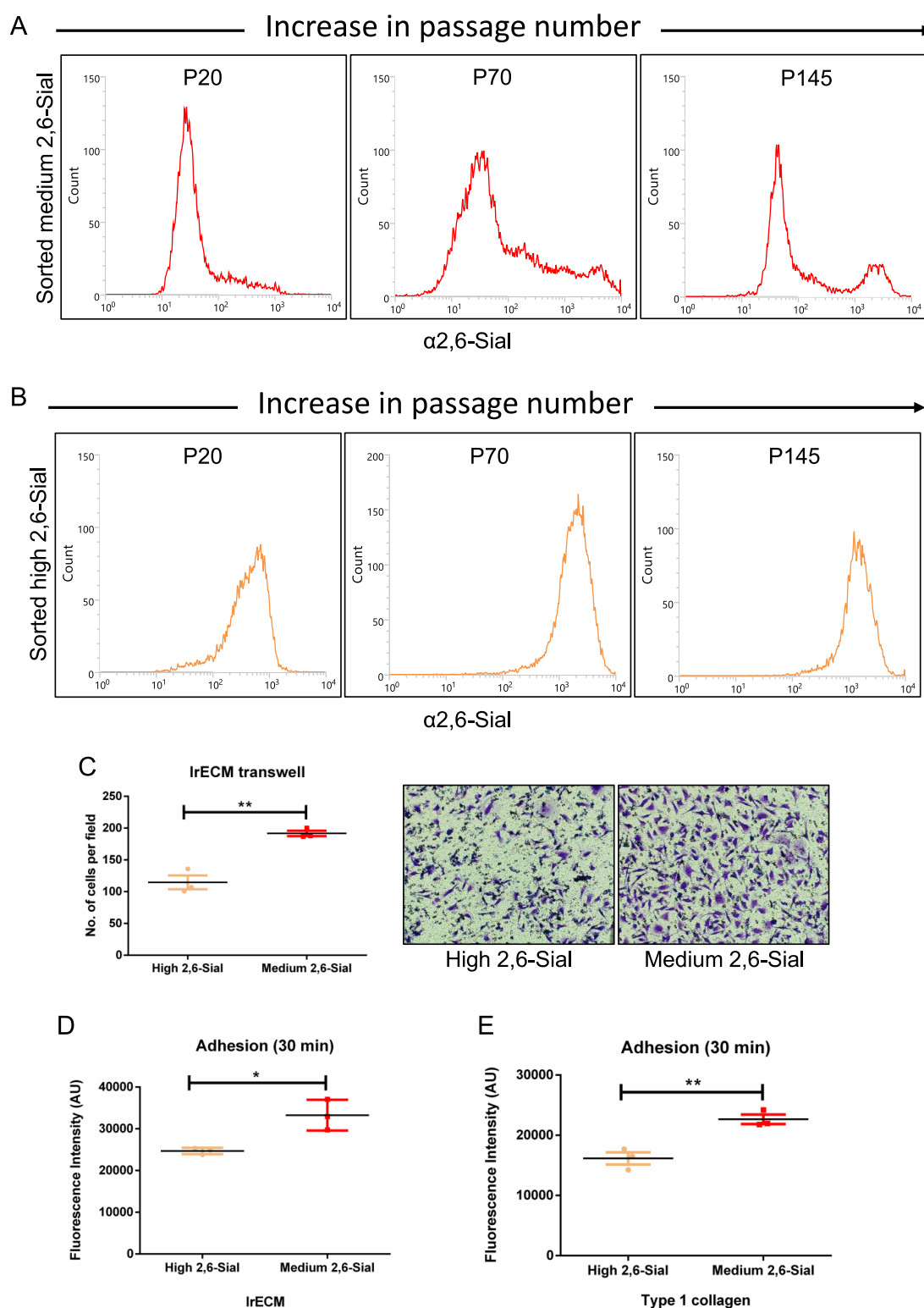


Figure 2. Medium 2,6-Sial cells show greater plasticity, adhere better to, and invade through ECM. (A) Lectin-based flow cytometry profile of $\alpha 2,6\text{-Sial}$ levels of medium 2,6-Sial population at three passages (20, 70, and 145) of long-term culture showing a gradual recapitulation of bimodal $\alpha 2,6\text{-Sial}$ distribution after 70 passages. (B) Lectin-based flow cytometry of high 2,6-Sial at three passages (20, 70, and 145) in long-term culture showing no change in $\alpha 2,6\text{-Sial}$ even after 145 passages. (C) Bar graph showing lower invasion of high 2,6-Sial cells (yellow) compared with medium 2,6-Sial cells (red) that invaded to the other side of IrECM-coated transwells ($n = 3$). (D) Graph showing lower adhesion of high 2,6-Sial cells (yellow) compared with medium 2,6-Sial cells (red) to IrECM ($n = 3$). (E) Graph showing lower adhesion of high 2,6-Sial cells (yellow) compared with medium 2,6-Sial cells (red) to Type 1 collagen ($n = 3$). Error bars denote mean $\pm$ SEM. The unpaired Student's t test was performed for statistical significance (* $P \leq 0.05$, ** $P \leq 0.01$).

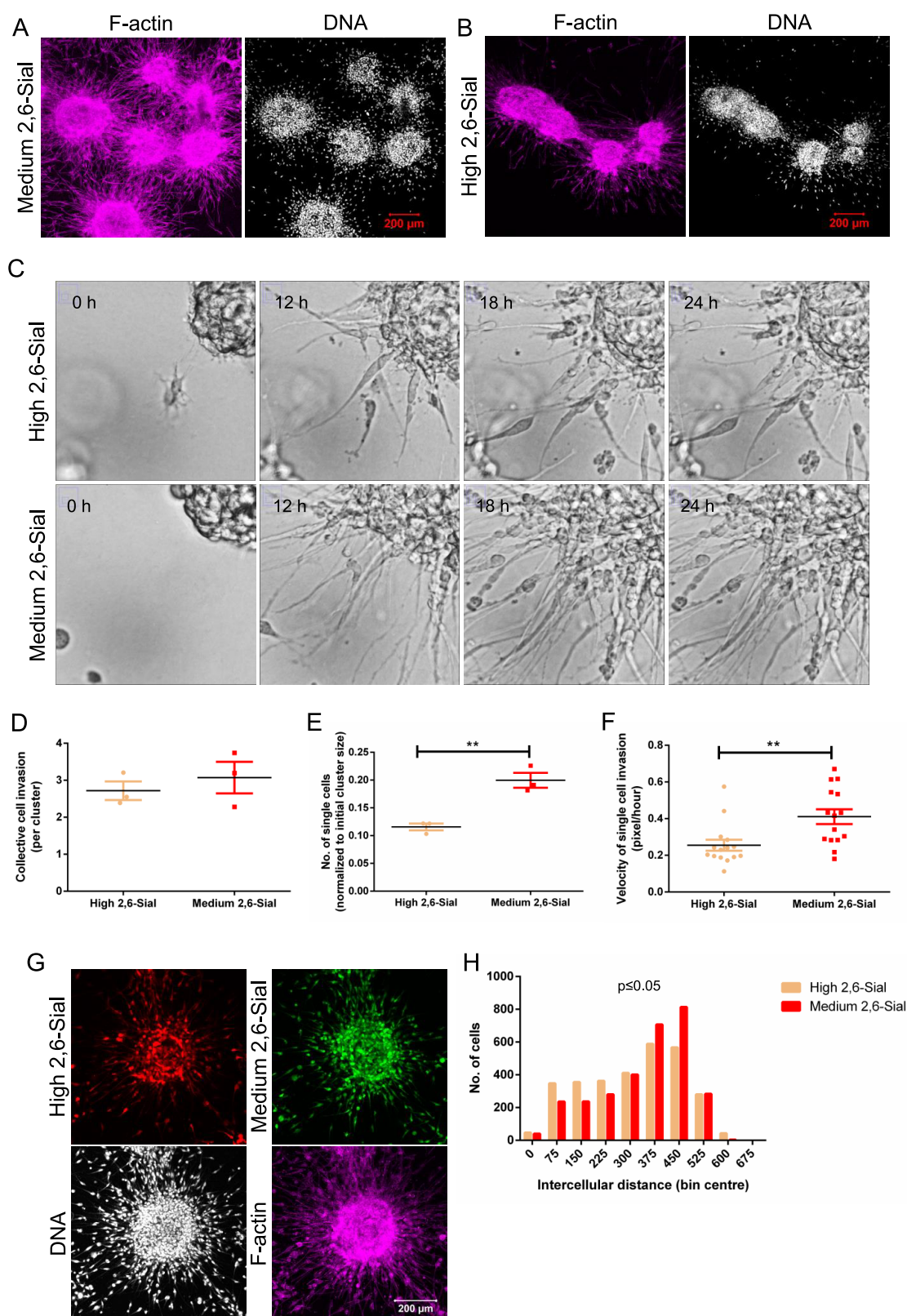


Figure 3. Medium 2,6-Sial cells invade faster through a 3D pathotypic multi-ECM microenvironment. (A) Confocal micrographs showing medium 2,6-Sial cells invading into fibrillar Type 1 collagen matrix from IrECM-coated multicellular clusters after 24 h. (B) Confocal micrographs showing high 2,6-Sial cells invading into fibrillar Type 1 collagen matrix from IrECM-coated multicellular clusters after 24 h. (A, B) Cells are counter-stained for the nucleus with DAPI (white) and F-actin with phalloidin (magenta) (scale bar: 200 μ m). (C) Bright field images taken at 0, 12, 18, and 24 h from time-lapse videography of IrECM-coated clusters of high and medium 2,6-Sial (top and bottom) invading into surrounding Type 1 collagen. (D) Graph showing insignificant differences in collective cell mode of invasion of high (yellow) and medium (red) 2,6-Sial cells as measured by the increase in cluster size obtained from time-lapse videography ($n = 3$). (E) Graph showing significantly lower invasion of high 2,6-Sial cells (yellow) compared with medium 2,6-Sial cells (red) as measured by the number of dispersed single cells in Type 1 collagen normalized to the initial cluster size obtained from

Figure 3. continued

lapse videography ($n = 3$). (F) Graph showing significantly lower mean migratory velocity of single high 2,6-Sial cells (yellow) compared with medium 2,6-Sial cells (red) as measured by manual tracking dynamics obtained from lapse videography ($n = 3$, $N \geq 15$ cells). (G) Confocal micrographs showing differential sorting of medium 2,6-Sial (green) cells invading and dispersing further into the Type 1 collagen ECM while high 2,6-Sial cells (red) form the core of the cluster when these two cells have been mixed in equal proportion and cultured in 3D. (Cells are counter-stained for the nucleus with DAPI (white) and F-actin with phalloidin (magenta) (scale bar: 200 μm). (H) Histogram showing distribution of intercellular distances between high 2,6-Sial cells (red) compared with medium 2,6-Sial cells (yellow). Intercellular distances between medium $\alpha 2,6$ -Sial cells are shifted rightwards indicative of a greater spread and farther invasion within 3D matrix microenvironment compared to high 2,6-Sial cells. Error bars denote mean $\pm$ SEM. The unpaired Student's t test was performed for statistical significance ($*P \leq 0.05$, $**P \leq 0.01$).

sialylated N -glycans with the terminal $\alpha 2,6$ linkage was higher in the high 2,6-Sial cells relative to medium 2,6-Sial cells, confirming observations of the lectin flow cytometry analysis. Interestingly, the core fucosylated versions of these isomers (m/z 1038.8²⁻ and m/z 1184.4²⁻) had a more pronounced difference in the $\alpha 2,3/6$ linkage relative abundance. The representative EICs of the sialylated structures (m/z 1131.4¹⁻, m/z 1040.3¹⁻, m/z 966.3¹⁻, and m/z 675.3¹⁻) from O -glycans (Figure S13, left) did not show any significant difference in their relative abundance of the sialylated species across the different cell types. Moreover, the detected sialylated residues linked to the galactose in the O -glycan analysis were all attributed to $\alpha 2,3$ linkages based on biological pathway processing, retention time, and negative mode fragmentation characterization from previous analysis.³⁰ The $\alpha 2,6$ -linked sialylated residues were found to be conjugated to the core GalNAc residue as observed for the structure of m/z 966.3¹⁻ and the early eluting isomer of m/z 675.3¹⁻. Similarly, the GSL sialylated glycans were also all $\alpha 2,3$ linked to the galactose as observed in structures with m/z 634.2¹⁻, m/z 925.3¹⁻, and m/z 999.3¹⁻. The glycan with m/z 925.3¹⁻ had an additional sialic acid found to be linked to the GM3 ganglioside structure and therefore attributed to the $\alpha 2,8$ linkage (Figure S13, right). These findings suggest that the intercellular heterogeneity that we observe for $\alpha 2,6$ -Sial is specific to sialylated N -linked glycans with little contribution from O -linked and GSL glycoconjugates.

Medium 2,6-Sial Cells Show Greater Plasticity and Invasion than High 2,6-Sial Cells. We next sought to investigate the functional differences between the high and medium 2,6-Sial subpopulations. Upon serial passaging in culture, medium 2,6-Sial cells showed a gradual recapitulation of the bimodal expression seen in flow cytometry of unsorted MDA-MB-231 cells (Figure 2A). On the other hand, the high 2,6-Sial cells consistently showed a high and unimodal level of $\alpha 2,6$ -Sial (Figure 2B). We did not observe any changes in $\alpha 2,3$ -Sial levels in both medium and high 2,6-Sial cells upon serial passaging (Figures S14 and S15).

Next, we investigated the ability of these subpopulations to invade through the ECM. Medium 2,6-Sial cells invaded more through lrECM [laminin-rich ECM also called as laminin-rich basement membrane (lrBM)]-coated transwells, when compared with high 2,6-Sial cells (Figure 2C). Invasion of mesenchymal cells requires strong tethering to ECM substrata.³² Therefore, the adhesion of the sorted populations to both lrBM and the fibrillar Type 1 collagen was assessed. For both matrices, higher adhesion was observed for medium 2,6-Sial cells when compared to high 2,6-Sial cells [Figure 2D,E; BSA-coated surface, as a negative control showing negligible cell adhesion (Figure S16)]. Higher adhesion to fibrillar Type 1 collagen was also observed for sorted BT-549 cells with lower 2,6-Sial levels (Figure S17). Furthermore, we assayed for adhesion of medium 2,6-Sial cells at various passages, namely,

early passage (P10–15), middle passage (P75–80), and late passage (P155–160). As the passage number of the cells increases, adhesion to both matrices decreased suggesting the role of $\alpha 2,6$ -Sial in cell–ECM adhesion (Figure S18). In addition, we assayed if there is any difference in proliferation rates of medium and high 2,6-Sial cells. We have not observed any significant difference in the number of viable cells at both 24 and 48 h between the populations (Figure S19).

Medium 2,6-Sial Cells Invade and Disperse Further than High 2,6-Sial Cells. The ability of medium 2,6-Sial cells to adhere better to, and invade more through, ECM than high 2,6-Sial cells led us to ask if the former invade in a collective manner or as single cells (referred to as mesenchymal invasion) wherein adhesion to ECM is crucial.³³ To answer the question, we used a customized 3D assay, wherein clusters of cancer cells are coated with lrBM matrix and then embedded within fibrillar Type 1 collagen to mimic the collagen-rich stromal environment.³⁴ In concurrence with our transwell experiments, single medium 2,6-Sial cells (Figure 3A) were found to radially invade into the Type 1 collagen to a greater extent than high 2,6-Sial cancer cells (Figure 3B). Time-lapse bright field microscopy on such cultures allowed us to measure the radial collective cellular migration as well as count the single cells that migrated into the fibrillar ECM (Figure 3C; Video S1A,B). Both medium and high 2,6-Sial cells showed comparable collective cell migration (Figure 3D). However, the number of single medium 2,6-Sial cells that invaded into the collagen as well as their mean migratory velocity were significantly higher than those of high 2,6-Sial cells (Figure 3E,F).

This prompted us to ask whether differential invasion would cause a population with both high and medium 2,6-Sial cells in a cancer cluster embedded in ECM to self-sort, with the former giving rise to the indolently growing bulk and the latter forming the radially invading front. This was indeed found to be the case (Figure 3G): when the two subpopulations were labeled with constitutively expressing fluorescence reporters, mixed in equal numbers and embedded in lrBM and Type 1 collagen, medium 2,6-Sial cells (Figure 3G, green) predominantly were present within the collagen, and high 2,6-Sial cells (Figure 3G, red) were closely clustered together in the central core. The relatively higher dispersion of the medium 2,6-Sial cells was confirmed by plotting the intercellular distances separately for medium and high 2,6-Sial cells wherein the histogram for the latter showed a leftward skew relative to the former (Figure 3H). Further, we asked if $\alpha 2,6$ -Sial levels in medium and high 2,6-Sial cells changed upon being cocultured with each other. Upon cocultivation of a 1:1 ratio of medium and high (RFP) 2,6-Sial cells for 48 h, we probed and did not observe any significant change in $\alpha 2,6$ -Sial expression levels for both the populations, indicating that the $\alpha 2,6$ -Sial of the two populations is retained in cocultures (Figure S20).

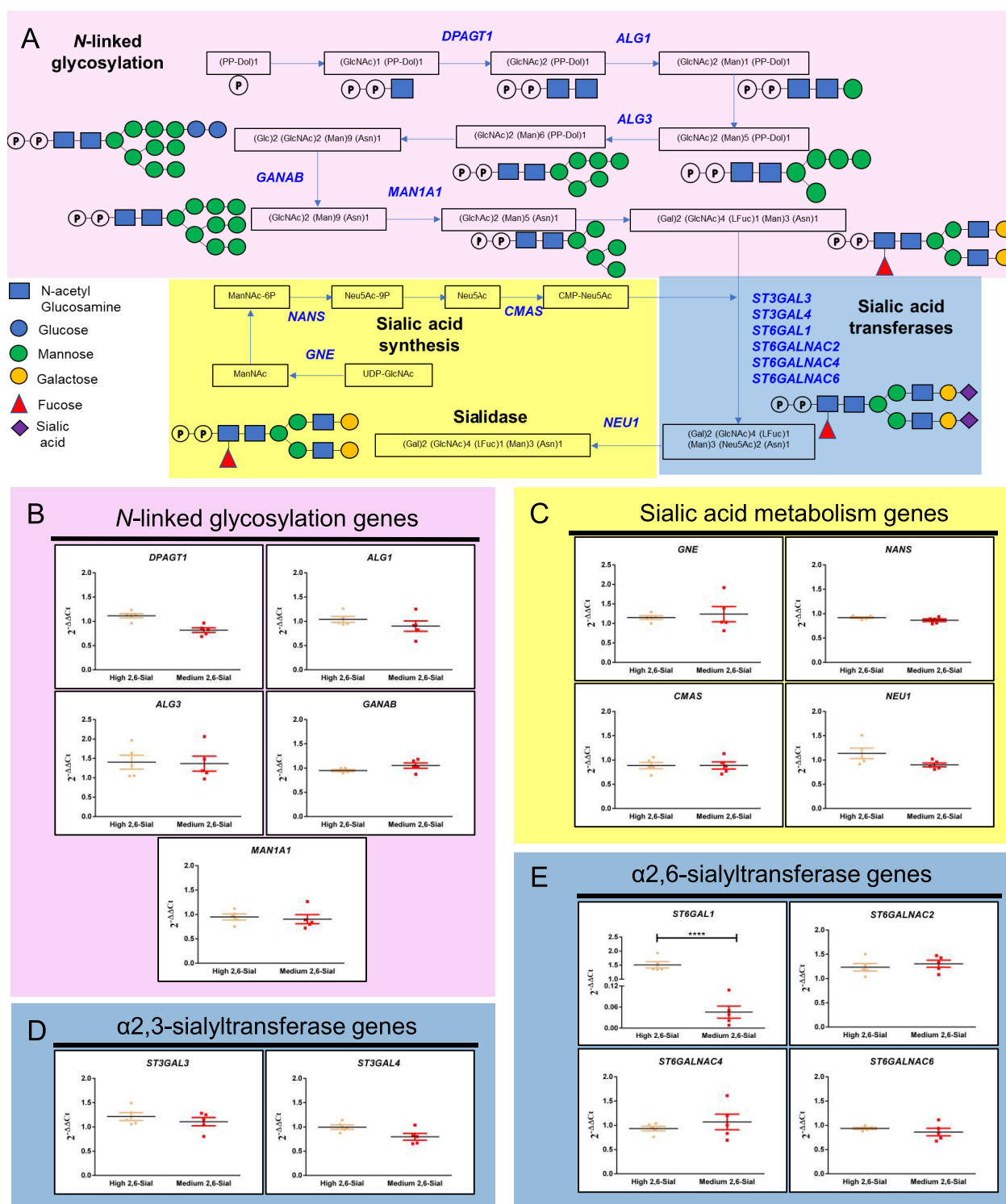


Figure 4. Differential expression of the *ST6GAL1* gene. (A) Schematic depiction of key processes involved in the fate and utilization of sialic acids and the genes encoding the enzymes involved in these processes arranged in the temporal order of their function: *N*-glycan synthesis, sialic acid metabolism and sialyltransferases. (B) Graphs depicting relative mRNA levels of genes involved in *N*-linked glycosylation *DPAGT1*, *ALG1*, *ALG3*, *GANAB*, and *MAN1A1*; expression is insignificantly altered between high (yellow) and medium (red) 2,6-Sial cells. (C) Graphs depicting relative mRNA levels of genes involved in sialic acid metabolism *GNE*, *NANS*, *CMAS*, and *NEU1*; expression is insignificantly altered between high (yellow) and medium (red) 2,6-Sial cells. (D) Graphs depicting relative mRNA levels of genes involved in 2,3-sialic acid conjugation *ST3GAL3* and *ST3GAL4*; expression is insignificantly altered between high (yellow) and medium (red) 2,6-Sial cells. (E) Graphs depicting relative mRNA levels of genes involved in 2,6-sialic acid conjugation *ST6GAL1*, *ST6GALNAC2*, *ST6GALNAC4*, and *ST6GALNAC6*; expression is insignificantly altered between high (yellow) and medium (red) 2,6-Sial cells for all except *ST6GAL1*, which is significantly lower in medium 2,6-Sial cells compared to high 2,6-Sial cells. Expression of all genes is plotted as $2^{-\Delta\Delta C_t}$ normalized to unsorted cells. The *18S* rRNA gene is used as an internal control. Data shown are from five independent biological experiments with at least duplicate samples run in each experiment. Error bars denote mean $\pm$ SEM. The unpaired Student's *t* test was performed for statistical significance ($****P < 0.0001$).

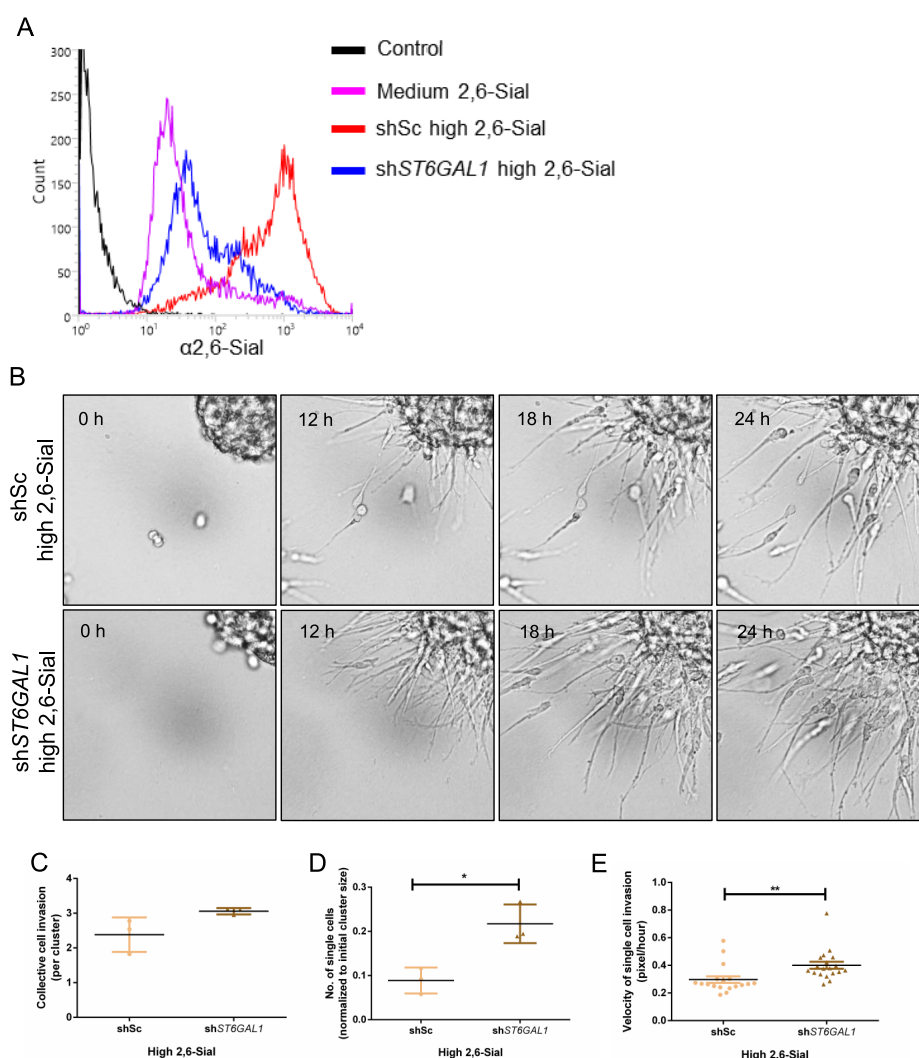


Figure 5. $\alpha 2,6$ -Sial levels regulate mesenchymal invasion of high 2,6-Sial cells. (A) Lectin-based flow cytometry profiles showing decreased $\alpha 2,6$ -Sial levels upon *ST6GAL1* gene knockdown in high 2,6-Sial cells. (B) Bright field images taken at 0, 12, 18, and 24 h from time-lapse videography of IrECM-coated clusters of scrambled control and *ST6GAL1* knocked down high 2,6-Sial cells (shSc-high 2,6-Sial top and shST6GAL1-high 2,6-Sial bottom) invading into surrounding Type 1 collagen. (C) Graph showing insignificant differences in the collective cell mode of invasion of shSc-high 2,6-Sial (yellow) and shST6GAL1-high 2,6-Sial (brown) cells as measured by the increase in cluster size obtained from time-lapse videography ($n = 3$). (D) Graph showing significantly lower invasion of shSc-high 2,6-Sial cells (yellow) compared with shST6GAL1-high 2,6-Sial cells (red) as measured by the number of dispersed single cells in Type 1 collagen normalized to the initial cluster size obtained from lapse videography ($n = 3$). (E) Graph showing significantly lower mean migratory velocity of single shSc-high 2,6-Sial cells (yellow) compared with shST6GAL1-high 2,6-Sial cells (red) as measured by manual tracking dynamics obtained from lapse videography ($n = 3$, $N \geq 15$ cells). Error bars denote mean $\pm$ SEM. The unpaired Student's t test was performed for statistical significance (* $P \leq 0.05$, ** $P \leq 0.01$).

Medium and High 2,6-Sial Cells Differ in Their Expression of *ST6GAL1*. We next asked whether the distinct $\alpha 2,6$ -Sial levels in the two sorted populations could be due to differential expression of genes involved in glycan synthesis and/or sialic acid metabolism. To do so, we examined the expression of genes encoding proteins involved in *N*-linked glycan synthesis (*ALG1*, *MAN1A1*, *DPAGT1*, *ALG3*, and *GANAB*) and sialic acid synthesis and sialidase (*CMAS*, *GNE*, *NANS*, and *NEU1*) and those coding for 2,3- and 2,6-sialyl transferases (Figure 4A) using quantitative real-time PCR (qRT-PCR). We did not observe any significant change in the expression of genes involved in *N*-linked glycosylation, sialic acid synthesis, and sialidase (Figure 4B,C) validating our flow cytometry results on the lack of detection of differences in *N*-linked glycans between the two sorted populations. The expression of genes encoding 2,3-sialyltransferases was not significantly changed confirming

the equivalent expression of 2,3-Sial levels in the medium and high 2,6-Sial cells (Figure 4D). We found that *ST6GAL1* mRNA levels were lower in medium 2,6-Sial cells when compared with high 2,6-Sial cells. In fact, with continuous passaging, as medium 2,6-Sial cells recovered the 2,6-Sial staining of the unsorted cells (Figure 2A), along with a concomitant decrease in ECM adhesion (Figure S18), we also observed a progressive increase in *ST6GAL1* expression (Figure S21). Expression levels of other $\alpha 2,6$ -sialyl transferase genes like *ST6GALNAC2*, *ST6GALNAC4*, and *ST6GALNAC6* did not vary significantly between populations (Figure 4E).

Knockdown of *ST6GAL1* in High 2,6-Sial Cells Enhances Their Mesenchymal Invasion. To establish if the $\alpha 2,6$ -sialic acid linkage has a direct effect on cancer cell invasion, we knocked down the expression of *ST6GAL1* in high 2,6-Sial cells using lentivirally delivered shRNA. *ST6GAL1*

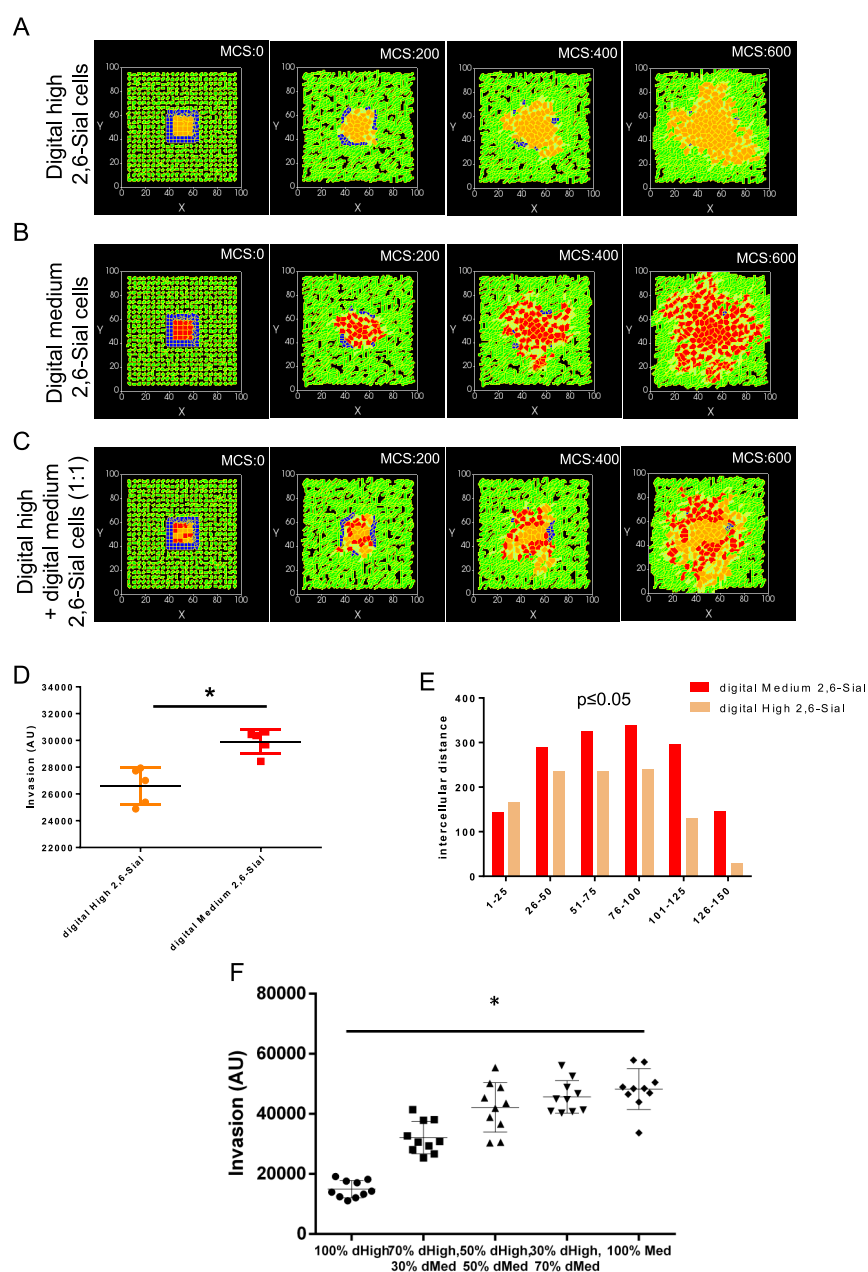


Figure 6. Multiscale simulations predict that matrix adhesion principally contributes to increased invasion. (A) Snapshots of simulation at MCS (0, 200, 400, and 600) of model lrECM-coated (blue) clusters of digital high 2,6-Sial cells in model Type 1 collagen (green). (B) Snapshots of the simulation at MCS (0, 200, 400, and 600) of model lrECM-coated (blue) clusters of digital medium 2,6-Sial cells in model Type 1 collagen (green). (C) Snapshots of simulation at MCS (0, 200, 400, and 600) of model lrECM-coated (blue) clusters of digital high and medium 2,6-Sial cells mixed in a ratio of 1:1 show a relatively greater invasion and dispersal of digital medium 2,6-Sial cells with digital high 2,6-Sial cells forming the central core. (D) Bar graph showing greater invasion of digital medium 2,6-Sial cells (red) compared with digital high 2,6-Sial cells in their 3D cocultures such as in Figure 6C ($n = 3$). Error bars denote mean $\pm$ SEM. The unpaired Student's t test was performed for statistical significance. (E) Histograms depicting the intercellular distances of the digital high 2,6-Sial cells (yellow) and those of medium 2,6-Sial cells (red) with the rightward shift of the latter indicating greater dispersal. (F) Graph depicting the invasion of an overall population of cancer cells in 3D (conditions similar to parts A–C) wherein the clusters of digital cells are composed of digital high and medium 2,6-Sial cells in the relative proportion of 100%, 0%; 75%, 25%; 50%, 50%; 25%, 75%; and 0%, 100%, from left to right, respectively. Error bars denote mean $\pm$ SEM. Ordinary one-way ANOVA with Tukey's post-hoc multiple comparisons was performed for statistical significance ($*P \leq 0.05$).

expression was confirmed using qRT-PCR (Figure S22). Lectin flow cytometry confirmed a resultant decrease of $\alpha 2,6$ -Sial levels in high 2,6-Sial cells with *ST6GAL1* knockdown (sh*ST6GAL1*-high 2,6-Sial cells) compared with scrambled control cells (shSc-high 2,6-Sial cells) (Figure 5A). As a result of the knockdown,

$\alpha 2,6$ -Sial levels of sh*ST6GAL1*-high 2,6-Sial cells were comparable to medium 2,6-Sial cells (Figure 5A).

When assayed for invasion from within lrBM to Type 1 collagen, sh*ST6GAL1*-high 2,6-Sial cells showed greater presence in collagen compared to shSc-high 2,6-Sial cells (Video S2A,B) (Figure 5B). Collective cell invasion did not

show any difference between knockdown and scrambled control cells (Figure 5C). However, the number of invaded single mesenchymal shST6GAL1-high 2,6-Sial cells in the collagen and their mean migratory velocity in ECM were greater than those of shSc-high 2,6-Sial cells (Figure 5D,E). These observations suggest that single cell invasion of breast cancer epithelia may be directly regulated by their surface α 2,6-Sial levels.

Multiscale Simulations Show That Altered Matrix-Adhesion Dynamics Is Sufficient to Explain Differential Invasion of Cancer Epithelia. Is the greater adhesion of medium 2,6-Sial cancer cells to ECM causal to its enhanced ability for invasion when cultured in 3D separately from, or in coculture with, high 2,6-Sial cells? To answer this, we employed a Cellular Potts-based computational model of cancer cell invasion³⁴ using the CompuCell 3D simulation environment.³⁵ Our model, calibrated to diverse matrix microenvironments and pharmacological perturbations, had earlier been shown to simulate single-cell and collective-cell invasion (individually and in combination, known as multiscale invasion) through consecutive barriers of lrBM-like collagen-like fibrillar environments, similar to the 3D invasion assay used above.³⁴ The cellular constituents of our model were digital medium and high 2,6-Sial cells (mimicking medium and high 2,6-Sial cells) with the former differing from the latter in exhibiting greater ECM adhesion. The ECM constituents of the model were digital lrBM and digital collagen (mimicking BM and Type 1 collagen) with the only difference being their nonfibrillar and fibrillar structure, respectively. In consonance with our experiments, digital medium 2,6-Sial cells showed greater invasion through model ECM than digital high 2,6-Sial cells (Figure 6A,B).

We next asked whether the computational model would predict the differential radial invasion that the medium 2,6-Sial cells showed, when cultured in an admixture with high 2,6-Sial cells. Even in our simulation framework, digital medium 2,6-Sial cells differentially migrated further to predominantly form the invasive front of mixed digital tumoroid populations, wherein digital high 2,6-Sial cells contributed to the core (Figure 6C; quantification of individual digital medium and high 2,6-Sial cell invasion when present within the initial mass in a ratio of 1:1 shown in Figure 6D and the further dispersion of digital medium 2,6-Sial cells confirmed through the intercellular distance histogram shown in Figure 6E). Given that the digital cells differ only with respect to their adhesion to the model ECM, the latter property is likely the determinant factor for the differential and greater invasion of real medium 2,6-Sial cells. Upon varying the ratio of digital medium and high 2,6-Sial cells constituting similarly sized initial clusters in our model, we observed that overall invasion was lowest in starting tumor clusters with 100% digital high 2,6-Sial cells (Figure 6F). A progressive increase in invasion was observed as the proportion of medium cells increased in the digital tumoroid mass, although no significant increase was seen between clusters with 50–100% medium-containing clusters. The distribution of invasion across 10 runs for these three conditions showed that the arrangement of cells within such clusters and the stochasticity of multiple interactions between cells and ECM during the simulation runs contributed to the magnitude of the phenotype. Notably, there were certain arrangements where clusters with as little as 50% digital medium cells “out-invaded” 100% digital medium 2,6-Sial cell clusters.

DISCUSSION

In this current study, we establish for the first time the occurrence of intercellular diversity in the expression of a

specific sialic acid linkage within malignant tumors and cancer cell lines of the breast. We show that cellular populations with distinct α 2,6-Sial levels coexist together within cell lines and may reflect linkage heterogeneity within tumor cell populations *in vivo*. The bimodal expression of α 2,6-Sial shown here can also be evidenced in several reports for MDA-MB-231 as well as other cell lines, but its consequence has not been investigated before.^{36–40} The α 2,6-Sial heterogeneity is counterpoised with relatively homogeneous expression in the same cells for other glycans such as α 2,3-Sial, fucose, T/Tn-antigen, bisecting/biantennary complex N-glycans, and tri/tetra antennary complex N-glycans. This suggests that glycan diversity within tumor populations may be specific to the identity of monosaccharides and their linkage to preceding glycan moieties.

Several published reports note hypersialylation of cancer cells, including that of the neoplasms of breast cancer.²² Hypersialylation is associated with, and suggested to be causal to, increased aggressiveness, stemness, resistance to chemotherapeutic agents, ability to survive in stressful conditions like hypoxia, and impaired nutrient supply.^{24,25,27,41} However, when we performed flow cytometry to isolate the two subpopulations showing distinct α 2,6-sialic acid levels, the one showing lower levels (which we denoted as medium 2,6-Sial cells) showed greater invasion than high 2,6-Sial cells both in the transwell assay and in 3D cultures. Medium 2,6-Sial cells were also observed to adhere better to both laminin-rich and collagenous ECM, as well as migrate through the latter with higher velocity compared to high 2,6-Sial cells. When cultured together, medium 2,6-Sial cells migrated farther and were dispersed to a greater extent than high 2,6-Sial cells. When α 2,6-sialic acid levels in high cells were genetically decreased to levels comparable with medium 2,6-Sial cells, their velocity and dispersal in ECM appropriately increased. Can the difference in invasion be explained by an appropriate difference in adhesion to ECM?

To answer this question, we used Cellular Potts model-based computational simulations calibrated with our experimental findings, which predicted that cells with higher adhesion to ECM are able to invade better into a surrounding stromalike fibrillar environment. Our simulations also were able to provide an answer to another question: what advantage do high 2,6-Sial cells confer to tumors in the presence of the more invasive medium 2,6-Sial cells? Simulations performed with different beginning ratios of digital medium and high 2,6-Sial cells showed that the presence of a population of slow invading high 2,6-Sial cells (that exhibit a relatively weaker adhesivity to surrounding ECM) could bias the migration of medium 2,6-Sial cells in a radially centrifugal manner. For specific initial conditions, such mixed cell populations could invade more than populations solely consisting of medium 2,6-Sial cells. The inertial behavior exhibited by high 2,6-Sial cells is prognostic of jamming–unjamming dynamics proposed to play an important role in the physical mechanisms of cancer progression and suggests that the sialic acid heterogeneity could give rise to heterogeneity in the material behavior of tumor subpopulations. In fact, investigating the interconversion of collective to mesenchymal migratory behavior, Ilina and co-workers have recently shown that the cell adhesion along with confinement by matrix may induce single-cell migratory behavior.⁴² Our results suggest a concert between the confining ECM and a jammed epithelial core in facilitating the mesenchymal migratory behavior of cancer epithelia. We will actively investigate this aspect in the future.⁴³

Our observations made through the assessment of endogenous expression of *ST6GAL1* (which is positively correlated with 2,6-Sial levels) serve to reconcile the contradiction between reports of increased invasion as a result of forced overexpression of *ST6GAL1* within cancer cells^{24,27} and an overall decreased level of *ST6GAL1* expression in breast cancer tissues assessed within The Cancer Genome Atlas.⁴⁴ We posit that, within a population with heterogeneous expression of α 2,6-sialic acids, those with a moderate expression will escape faster through a mesenchymal invasive process. At the same time, the medium 2,6-Sial cells will continue to give rise to the high 2,6-Sial populations, as demonstrated by our flow cytometric experiments on continuously passaged sorted cells. In this study, we have not investigated whether there are differences in the set of cell surface proteins to which the 2,6-sialic acids are conjugated, between the sorted populations. The deployment, in the future, of synthetic sialylated glycopolymers on cancer cell surfaces and their effects on cell adhesion could be used to ascertain whether the effect of sialic acids on adhesion is a direct one or based on downstream signaling. In conclusion, our results show not only that the intercellular sialic acid heterogeneity and breast cancer cell invasion are spatiotemporally coincident but also that these two processes actively drive each other. It will be imperative to break the link between the two by targeting sialic acids through novel therapeutic strategies such as precision glyco-editing⁴⁵ and sialic acid–siglec-based immunotherapy.^{46,47}

MATERIALS AND METHODS

Cell Culture. MDA-MB-231 cells were maintained in DMEM:F12 (1:1) (HiMedia AT140) along with 10% fetal bovine serum (Gibco, 10270) in a 5% carbon dioxide, 37 °C temperature humidified incubator.

Lectin Histochemistry. Breast tumor and normal sections were made from paraffin-embedded blocks at Kidwai cancer institute, Bangalore, after obtaining necessary approval from the Institutional Human Ethics committee and consent from patients. Sections were incubated at 65 °C overnight to remove wax. Immediately, samples were rehydrated gradually incubating in decreasing concentrations of alcohol: 2 × 5 min Xylene, 2 × 5 min 100% ethanol, 2 × 5 min 90% ethanol, 1 × 10 min 80% ethanol, and 1 × 10 min 70% ethanol and finally in distilled water for 10 min. Antigen retrieval was performed using citrate buffer pH 6.0 in the microwave for 30 min and allowed to cool down to room temperature. Sections were blocked using 1× Carbo-Free blocking buffer (Vector laboratories, SP-5040) made in PBS pH 7.4 for 1 h at room temperature. Fluorescently labeled SNA (Vector Laboratories, FL-1301) and MAA (bioWORLD, 21511106-1) were added to sections at 1:100 dilution and incubated overnight at 4 °C. Lectins preincubated with 250 mM lactose (HiMedia, RM565G) were used as a negative control. Sections were washed with 1× PBS for 5 min at room temperature thrice. Sections were counterstained with 1 μ g/mL DAPI (Thermo Fisher Scientific, D1306) for 10 min, washed to remove excess stain, and mounted for imaging. CellProfiler software was used for per cell quantification. A pipeline was built where the nucleus, α 2,6- and α 2,3-Sial channels were separated. The cell nuclei mask was obtained using a thresholding for the diameter of the nucleus between 20 and 35. Particles outside this diameter range and near the border of the image were discarded. Clumps of the nucleus were resolved using the shape algorithm available in the software. Such obtained nucleus masks were overlaid onto the α 2,6-Sial/

α 2,3-Sial image channel, and integrated intensity per cell was calculated.

Lectin Cytochemistry. Cells (15 000) were seeded in an 8-well chamber cover glass (Eppendorf, 0030742036). After 24 h, spent medium was removed and cells were washed with cold 1× PBS once and fixed using 4% formaldehyde at 4 °C for 20 min. Excess fixative was removed and neutralized with 2% glycine for 30 min at room temperature. Cells were washed thrice with 1× PBS and blocked using 1× Carbo-Free blocking buffer for 1 h at room temperature. Fluorescent conjugated SNA and MAA were added to cells at 1:500 dilution in blocking buffer and incubated for 3 h at room temperature or overnight at 4 °C. Cells were with 1× PBS for 5 min thrice, counterstained with 1 μ g/mL DAPI and 1:500 Alexa FluorTM633-conjugated phalloidin (Thermo Fisher Scientific, A22284) for 1 h at room temperature. Finally, cells were washed with 1× PBS for 5 min twice and imaged.

Lectin Flow Cytometry and Sorting. MDA-MB-231 cells were trypsinized and counted. 0.3×10^6 cells for analysis and 10^6 cells for sorting were taken in a polypropylene FACS tube in 100 and 500 μ L for analysis and sorting, respectively, in 1× Carbo-Free blocking buffer. Cells were incubated with fluorescently labeled SNA and MAA at a 20 μ g per 10^6 cells concentration for 20 min at room temperature. Lectin incubated with 250 mM lactose overnight at 4 °C overnight or FITC was used as the negative control. Finally, cells were diluted to 10^6 /mL using Carbo-Free blocking buffer and analyzed/sorted using a BD influx flow cytometer. For analysis, at least 10 000 total events were acquired. For sorting, single-cell purity mode was used. Lectins used in this study are listed in Table S1. The following gating strategy was employed to analyze the flow cytometry data. Briefly, a total of 10 000 events were recorded with forward scatter (FSC) on the X-axis (linear scale) and side scatter (SSC) on the Y-axis (log scale). Potential cells were gated and labeled as P1. To obtain single cells, cells from P1 were plotted on another dot plot, where the X-axis represented the forward scatter area (FSC-A), and the Y-axis represented forward scatter (FSC) both in the linear scale. Cells from P1 present along the diagonal were gated and labeled as P2. For further analysis, cells from only P2 were considered.

ECM Coating for the Adhesion Assay. 96-well plates were coated with 50 μ g/mL reconstituted basement membrane (rBM or IrBM) (Corning, 354230) or 50 μ g/mL neutralized rat tail collagen (rich in Type 1 collagen) (Gibco, A1048301) for 2 h at 37 °C. Excess matrix was removed, allowed to dry for 30 min at 37 °C, and blocked with 0.5% BSA (HiMedia, MB083) overnight at 37 °C. After overnight blocking, excess BSA was removed, and plates were used for the adhesion assay. The 0.5% BSA overnight coating at 37 °C was used as a negative control.

Adhesion Assay. MDA-MB-231 high and medium α 2,6-sialic acid cells were trypsinized. After counting, 30 000 cells per well were incubated in BSA- and ECM-coated wells for 30 min at 37 °C. Unadhered cells were removed carefully, and wells were washed with 1× phosphate buffered saline pH 7.4 (PBS) thrice to remove unadhered cells. Cells were fixed using 100% methanol for 10 min at room temperature. After fixing, cells were washed with 1× PBS thrice, stained with 50 μ g/mL propidium iodide (HiMedia, TC252) for 30 min at room temperature, and washed thrice with 1× PBS. Using the plate reader, fluorescence was read at Ex 535 nm/Em 617 nm. BSA or ECM without cells was used as the blank. The assay was done in triplicates and repeated three times independently.

Quantitative Real-Time PCR. Total RNA was isolated from high and medium α 2,6-Sial cells using TRIzol as per the

manufacturer's protocol (Invitrogen, 15596078). Total RNA was quantified using a UV–vis spectrophotometer (NanoDrop, Thermo Fisher Scientific). Total RNA (1 μ g) was reverse transcribed using a Verso cDNA synthesis kit as per the manufacturer's protocol (Thermo Scientific, AB-1453). Real-time PCR was performed with 1:2 diluted cDNA using a SYBR green detection system (Thermo Fisher Scientific, F415L) and Rotorgene Q (Qiagen, 9001560). The 18S rRNA gene was used as the internal control for normalization. Relative gene expression was calculated using the comparative Ct method, and gene expression was normalized to unsorted cells. All the genes analyzed along with sequence are mentioned in Table S2. Appropriate no template and no-RT control were included in each experiment. All the samples were analyzed in duplicates/triplicates and repeated three times independently.

Genetic Perturbation of the ST6GAL1 Gene. The ST6GAL1 gene shRNA clone was obtained from the MISSION shRNA library (Sigma Merck). Plasmid containing shRNA or scrambled control was packaged into lentivirus using packaging vectors pMD2.G and psPAX2 (packaging vectors were a kind gift from Dr. Deepak K Saini). The plasmids were transfected into 293FT cells (Thermo Fisher Scientific, R70007) using TurboFect (Thermo Fisher Scientific, R0533). Cells were cultured in DMEM supplemented with 10% FBS; conditioned medium containing viral particles was collected at 48 and 72 h. After filtering through a 0.45 μ m filter, viral particles were concentrated using the Lenti-X concentrator as mentioned in the manufacturer's protocol (TaKaRa, 631232). Concentrated virus was aliquoted and stored at -80°C until use. High α 2,6-Sial cells were seeded in a 24-well plate at 50–60% confluence and transduced with viral particles containing shRNA or scrambled control along with polybrene (4 μ g/mL) for 24 h. After 72 h, transduced cells were selected using 5 μ g/mL puromycin (HiMedia, CMS8861). The knockdown of the gene was assayed using real-time PCR and lectin flow cytometry as described earlier.

Transwell Invasion Assay. Polycarbonate transwell inserts (8 μ m pore-size) were obtained from HiMedia (TCP083). Transwells were coated with 200 μ g/mL reconstituted basement membrane (rBM or IrBM) (Corning, 354230) as per the manufacturer's protocol. In each transwell, 3×10^4 cells were seeded in 200 μ L of serum-free DMEM:F12 (1:1) medium. The bottom well was filled with 1 mL of 10% serum containing DMEM:F12 (1:1) and incubated for 24 h at 37°C , in a 5% carbon dioxide-containing humidified chamber. Carefully, medium from the transwell was removed and washed with $1\times$ PBS once, and cells were fixed using 100% methanol for 10 min. After fixing, cells were washed once again with $1\times$ PBS, and noninvading cells were carefully removed with a moistened cotton swab. Transwells were stained with 1% crystal violet for 15 min and washed to remove excess dye. Membranes were dried and imaged under a microscope using $40\times$ total magnification. At least 5 independent fields were imaged per transwell, and the numbers of cells were counted. Each experiment has been performed in duplicates and repeated three times.

3D Invasion Assay. A 3D invasion assay was performed as described previously by our group.³⁴ Briefly, cancer clusters were made using 30 000 cells in a polyHEMA-coated (Sigma, P3932), 96-well plate, defined medium⁴⁸ supplemented with 4% rBM. After 48 h, clusters were collected and embedded in polymerizing rat tail collagen in a chambered cover glass. 3D cultures were grown for 24 h in a 37°C humidified incubator

with 5% carbon dioxide. End point imaging was done after fixing 3D cancer clusters, counterstained with DAPI, Alexa FluorTM 633–phalloidin, and imaged using a Carl Zeiss LSM880 confocal microscope with system optimized settings. Bright field time-lapse imaging of invading cancer clusters was performed on an Olympus IX73 fluorescence microscope fitted with a stage top incubator and 5% carbon dioxide. Images were collected for 24 h with every 10 min interval.

Glycan Extraction and Mass Spectrometry. Glycan extraction and enzymatic release of glycans from proteins and lipids were slightly modified from previous protocols.^{30,31,49} Cells were harvested with a nonenzymatic method, such as scraping from plates using cell dissociation buffer (13151014, Gibco) for 5–10 min. A $1\times$ lysis solution was prepared using RIPA buffer (Genesearch; cat. no. 9806S), and the Roche complete protease inhibitor cocktail (CO-RO, Roche; Merck 11697498001) was used. 500 μ L of RIPA buffer was added to each tube and left at 4°C overnight. Samples were sonicated for 10 min in a sonication bath. Then, they were centrifuged for 10 min at 16 000g, and the supernatant was collected in new a microfuge tube. A triphasic liquid of chloroform:methanol:water (1:2:2) was used to separate the lipids, glycolipids, and proteins. Samples were vortexed and left on ice for 1 h. The samples were then centrifuged for 10 min at 10 000 rpm using a table-top centrifuge. The aqueous top layer containing the glycolipids was separated, and then, two volumes of methanol were added to each sample and vortexed to precipitate the proteins. Then, samples were centrifuged for 10 min at 10 000 rpm. The top layer (lipids) was carefully removed, while the pellet was resuspended in 100 μ L of 4 M urea and 3% SDS and quantified using the BCA assay. Protein lysate (20 μ g) from each of the replicates was independently dot blotted onto EtOH-wetted PVDF membrane (20 μ g of BSA and bovine fetuin were included as the glycan negative and positive controls, respectively) and dried overnight (RT). Dried dot blots were then washed in MeOH (15 min, shaking, RT) and further washed in water (15 min, shaking, RT). To qualitatively confirm and visualize the immobilized protein, the dot blotted membrane was stained with direct blue staining solution for 3 min and then destained, and the membrane was placed in water. Protein-laden spots of 6 mm diameter were excised from the PVDF membrane and submerged into wells containing 100 μ L of 1% PVP40 where they were incubated for 5 min and then washed thrice with 200 μ L of water. For the removal of N-linked glycans, 2 μ L of PNGase F (1000 U) (NEB) and 8 μ L of water were added to each well and incubated overnight at 37°C . After incubating overnight, the 96-well plate was sonicated for 5 min in an ultrasonic cleaner sonicating water bath (Unisonics). The wells were then rinsed with 20 μ L of water and the solutions collected. The samples were then acidified with 10 μ L of 100 mM ammonium acetate (pH 5) and incubated for 1 h, RT. Released N-glycans were reduced in alkaline conditions by the addition of 20 μ L of 1.25 M NaBH₄ in 100 mM KOH and incubated for 3 h at 50°C . After cooling to RT, the reaction was neutralized with 2 μ L of glacial acetic acid.

Sodium salts of N-glycan solutions were removed by passage through and subsequent washing of cation exchange micro-columns constituted by packing 25 μ L of Dowex 50W X8 (Sigma-Aldrich) in a ZipTip C18 tip (Merck Millipore), activated with 50 μ L of 1 M HCl prior to the addition of sample. Solutions were passed through columns using a benchtop microcentrifuge at full speed for 15–30 s. Sodium-desalted samples were vacuum-dried in a Savant SPD131DDA

SpeedVacTM concentrator (ThermoFisher). Samples were then washed and dried three times with 100 μ L of methanol to remove residual borate.

Desalted samples were further purified by porous graphitized carbon (PGC) chromatography. Columns were constituted with 5 μ L of PGC material from HyperSep Hypercarb cartridges (ThermoFisher) in methanol and deposited in a ZipTip C18 tip. Prior to the addition of the sample, columns were washed with elution buffer [80% v/v acetonitrile (ACN), 0.1% v/v trifluoroacetic acid (TFA)] and subsequently equilibrated in loading buffer (0.1% v/v TFA). Dried sample was dissolved in 50 μ L of loading buffer and passed through columns, with flow through reloaded into the columns. Enriched glycans were eluted in 50 μ L of elution buffer and vacuum-dried. Dried, purified enriched glycans were stored at -20°C until analysis using MS.

The O-glycans were sequentially released from the PVDF blotted proteins after the N-glycan release by PNGase-F by chemical beta-elimination. The immobilized proteins were subjected to high-alkaline conditions by the addition of 30 μ L of 0.5 M NaBH₄ in 100 mM KOH and incubated for 16 h at 50°C . After cooling to RT, the reaction was neutralized with 2 μ L of glacial acetic acid. The O-glycans were desalted and purified identically as described for the N-glycans.

Similarly, to release the glycans linked to glycolipids, the glycolipid-containing aqueous fraction was evaporated to dryness. The extract was redissolved in 100 μ L of 50 mM sodium acetate containing 0.1% sodium taurodeoxycholate, and 2 μ L (4 mU) of endoglycoceramidase II (Sigma) was added to each sample followed by 16 h of incubation at 37°C . Subsequently, 1 mL of chloroform/methanol/water (8:4:3, v/v/v) was added to each sample. The upper methanol:water layer containing the released oligosaccharides was transferred into a new microfuge tube and dried. The released glycans were reduced to alditols identical to the N-glycan method described above. Finally, purified glycan alditols were resuspended in 30 μ L of water prior to MS analysis. Sample handling and injections were performed using an Ultimate 3000 UHPLC instrument (Thermo Scientific). Samples were injected in loading buffer (10 mM NH₄HCO₃) through a PGC precolumn (3 μ m Hypercarb, 320 μ m ID $\times$ 100 mm) at a flow rate of 6 μ L/min and subsequently a PGC analytical column (3 μ m Hypercarb, 75 μ m ID $\times$ 100 mm) at a flow rate of 300 nL/min. Chromatographic separation was achieved using an 85 min gradient for N-glycans (0–70% v/v ACN) and 75 min (0–70% v/v ACN) gradient for O-linked and GSL-glycans. The eluting glycans were detected using an amaZon ETD speed ion trap (Bruker, Bremen, Germany) in the negative ion mode using a captive spray source. An m/z range 320–1500 was fixed for data-dependent precursor scanning. Both MS and MS/MS data were recorded in the instrument's ultrascan mode with an ICC target of 30 000 and accumulation time of 200 ms. Capillary exit voltage was set to 140 V, dry gas temperature at 77°C , and flow rate at 300 L/min. HV capillary voltage was set to 1300 V and the HV end plate offset to 500 V. Collision induced dissociation (CID) fragmentation was performed on the five most intense precursors of each MS scan.

Data analyses were carried out in Compass Data Analysis 4.2 (Bruker Corporation) for structural assignment and in Skyline 20.1 for quantitation. Quantitation was performed for each isomer identified by MS² spectra and retention time by calculating the area under the curve (AUC).

Compucell 3D Model and MATLAB Analysis. The simulations performed were based on an earlier-established computational model in Compucell3D.³⁴ Compucell3D (CC3D) is an environment that comprises the lattice-based GGH model [GGH: Glazier–Graner–Hogeweg also known as the Cellular Potts model (CPM)] with solvers for partial differential equations in order to simulate biological processes, wherein molecules, cells, and cellular ensembles may behave concurrently across distinct spatial scales.^{35,50} The software divides the whole simulation lattice into “cells” (collection of pixels). A specific “cell type” is assigned to each of them. All cells having the same cell type share the same properties. Hence, the minimalistic target tissue-system can be broken down into cell types representing the main constituents of the system. Interaction parameters between cell types can be made to approximate biological constraints between components, similar to that of the original *in vitro* or *in vivo* biological system. Such constraints or parameters regulate the simulation through the effective energy or Hamiltonian (H) calculated at each Monte Carlo step (MCS). The initial conditions and the behavior of cells in the model have been calibrated based on the invasive behavior of cancer cells in different types of ECM and upon treatment with pharmacological agents. The chief constituents of the model include cancer cells, nonfibrillar BM matrix, fibrillar Type 1 collagen matrix, degraded and newly synthesized cancer ECM, the diffusible activator of ECM (such as MMPs), and its diffusible inhibitor (TIMP).³⁴ Contact energies related to adhesion between each of these constituents and diffusion and cooperative interaction between the molecules determine the behavior of the simulation and the end stage phenotype and are based on calibrations with experiments involving tumoroids cultured in diverse matrix microenvironments and pharmacological perturbations (Pally et al, 2019). In this paper, we implemented an additional chemotaxis plugin to be able to simulate the movement of cells in response to attractant cues of the surrounding fibrillar ECM. Several parameter values, such as contact energy between the “cell” and “medium”, and diffusion coefficients of ECM proteolysis regulators were taken from previous published efforts by other computational groups that studied cancer cell migration through the ECM in order to conceptually integrate our model with the existent relevant literature.^{51–54}

Key differences with the previously mentioned version of the model are as follows:

- (1) Simulated cancer cells have two cell types assigned to them. They correspond to high and medium 2,6-Sial cells in the heterogeneous population.
- (2) The two different cancer cells have differential adhesion to ECM, where medium 2,6-Sial cells have relatively higher adhesion (1.5-fold based on our experimental findings) to all ECM cells like BM (basement membrane, bloblike), Type 1 collagen (collagen 1, fibrillar), newly synthesized collagen-like ECM, and degraded ECM. Widening or narrowing the fold change in ECM adhesion could lead to dramatic or subtler alterations in sorting dynamics between the differentially adhering cells, respectively. All other biological cancer cell-like properties are the same for the digital high and medium 2,6-Sial cells.

At the initial spatial configuration (marked by MCS 0), both digital high and medium 2,6-Sial cells are located centrally encapsulated by BM cells, and that is further surrounded by

Type 1 collagen cells. After the Monte Carlo step (MCS) reaches 600, the cancer cells already invade the matrix with reconstitution of the localized matrix through reaction-diffusion dynamics of MMPs and their inhibitors such as TIMPs. An image of the simulation at MCS 600 is collected for analysis in MATLAB.

Two different methods of analysis were used; the first one, “Area of minimal enclosing circle”, detailed in the previous publication³⁴ calculates the smallest possible circle, which encloses all cells of a certain cell type and can be considered as a quantification of invasion corresponding to that cell type in that initial configuration. The collected simulation images can be binarized for digital high 2,6-Sial cells (orange), digital medium 2,6-Sial cells (red), or both (orange and red; Figure 6F) depending on the analysis. The second method calculates distances between cells in simulation images binarized with respect to either high or medium cell type. After identifying centroids of the cells in the binarized image, a MATLAB function “pdist” was used to calculate distances between all the pairs of centroids and plotted in a histogram (Figure 6E) [<https://in.mathworks.com/help/stats/pdist.html>].

Statistical Analysis. All experiments were performed in duplicates or more. All experiments were repeated thrice independently. Prism software (GraphPad Prism 6.0) was used for the generation of graphs and analysis. For all experiments, results are represented as mean $\pm$ SEM unless mentioned. For statistical analysis, an unpaired, two-tailed Student's *t* test or ordinary one-way ANOVA followed by post hoc Tukey test for the comparison of multiple groups was performed. Significance (*p* value) is represented as *, where * $\leq$ 0.05, ** $\leq$ 0.01, *** $\leq$ 0.001, and **** $<$ 0.0001.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscentsci.0c00601>.

Negative/sugar control for lectin histochemistry, lectin cytochemistry, lectin-based flow cytometry, gating of single cells in flow cytometry, the specificity of SNA and MAA lectins, α 2,6- and α 2,3- Sial linkage profiles and adhesion of BT-549 to type 1 collagen; extracted ion chromatograms for *N*-glycans, *O*-glycans, and glycosphingolipids; and lectins and primers used in the study (PDF) Video S1: Time-lapse videography of high 2,6-Sial cells (MP4)

Video S2: Time-lapse videography of medium 2,6-Sial cells (MP4)

Video S3: Time-lapse videography of shSc-high 2,6-Sial cells (MP4)

Video S4: Time-lapse videography of shST6GAL1-high 2,6-Sial cells (MP4)

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Author Contributions

D. Pally, A. Everest-Dass, and R. Bhat designed the experiments. D. Pramanik and R. Bhat designed the simulations. D. Pally, A. Everest-Dass, S. Hussain, S. Verma, and A. Srinivas performed the experiments. D. Pramanik performed the simulations. D. Pally, A. Everest-Dass, D. Pramanik, and R. Bhat analyzed the data. D. Pally, A. Everest-Dass, R. Bhat, and R. V. Kumar wrote the manuscript.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank the Bioimaging facility at Division of Biological Science, Indian Institute of Science for help with confocal microscopy. We thank Bhaskar Kumawat for help with biostatistics. D. Pally is supported by Senior Research Fellowship (SRF) from MHRD, India. This work was supported by the Wellcome Trust/DBT India Alliance Fellowship/Grant [grant number IA/I/17/2/ 503312] awarded to R.B. In addition, this work was supported by funds from the Department of Biotechnology, India [BT/ 909 PR26526/GET/119/92/2017], SERB[ECR/2015/000280], DBT-IISc partnership program (BT/PR27952/INF/22/212/2018), and the Institute of Eminence grant (IE/CARE-19-0319). D. Pramanik acknowledges KVPY for the scholarship.

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